SIMATS ENGINEERING
ASSESSMENT TOOL 2

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REG NO: 192
COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.

4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.

5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.

4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1.Smart Campus Network (192.168.100.0/24)

School of Computing (/25)
Step 1: Subnet Mask = 255.255.255.128
Step 2: Block Size = 128
Network Address: 192.168.100.0
Host IP Range: 192.168.100.1 – 192.168.100.126
Broadcast Address: 192.168.100.127
Usable Hosts: 2 − 2 = 126

School of Electronics (/26)
Step 1: Subnet Mask = 255.255.255.192
Step 2: Block Size = 64
Network Address: 192.168.100.128
Host IP Range: 192.168.100.129 – 192.168.100.190
Broadcast Address: 192.168.100.191
Usable Hosts: 2 − 2 = 62

School of Mechanical Engineering (/27)
Step 1: Subnet Mask = 255.255.255.224
Step 2: Block Size = 32
Network Address: 192.168.100.192
Host IP Range: 192.168.100.193 – 192.168.100.222
Broadcast Address: 192.168.100.223
Usable Hosts: 2 − 2 = 30

Remaining Unused IP Address Range
Unused Range: 192.168.100.224 – 192.168.100.255

2. Multinational Software Company (10.10.0.0/16)
Software Development (/24)

Step 1: Subnet Mask = 255.255.255.0
Network Address: 10.10.0.0
Host IP Range: 10.10.0.1 – 10.10.0.254
Broadcast Address: 10.10.0.255
Usable Hosts: 2 − 2 = 254

Quality Assurance (/25)

Step 1: Subnet Mask = 255.255.255.128
Network Address: 10.10.1.0
Host IP Range: 10.10.1.1 – 10.10.1.126
Broadcast Address: 10.10.1.127
Usable Hosts: 126

Human Resources (/26)
Step 1: Subnet Mask = 255.255.255.192
Network Address: 10.10.1.128
Host IP Range: 10.10.1.129 – 10.10.1.190
Broadcast Address: 10.10.1.191
Usable Hosts: 62

Executive Management (/27)
Step 1: Subnet Mask = 255.255.255.224
Network Address: 10.10.1.192
Host IP Range: 10.10.1.193 – 10.10.1.222
Broadcast Address: 10.10.1.223
Usable Hosts: 30

Remaining Unused IP Address Range
Unused Range: 10.10.1.224 – 10.10.255.255

3. IPv6 Address

Given Address:
2001:0db8:abcd:0000:0000:0000:0000:0000/64

Step 1: Compress the IPv6 Address
Remove leading zeros and replace consecutive zero blocks with "::".
Compressed Address:
2001:db8:abcd::/64

Step 2: Expand the Address
Expanded Address:
2001:0db8:abcd:0000:0000:0000:0000:0000

Step 3: Network Prefix (/64)
The first four groups form the network prefix.

Network Prefix:
2001:db8:abcd:0::/64

Step 4: Total Host Addresses
Host bits = 128 − 64 = 64
Total Hosts = 2
= 18,446,744,073,709,551,616

Step 5: Total Possible Host Addresses
Answer: 18,446,744,073,709,551,616 host addresses

5. **Routing Table**

Given Routing Table

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination IP: 192.168.2.45
Step 1: Destination Subnet
The IP 192.168.2.45 belongs to
Subnet: 192.168.2.0/24

Step 2: Matching Routing Entry
192.168.2.0/24

Step 3: Next-Hop Network/Interface
Forward the packet through the router interface connected to the 192.168.2.0/24 network

Step 4: Default Route
If no route matches, use the
Default Route: 0.0.0.0/0

5. Administration LAN and Finance LAN
Administration LAN (192.168.10.0/26)

Step 1: Network and Broadcast Address
Network Address: 192.168.10.0
Broadcast Address: 192.168.10.63

Step 2: Host IP Range
192.168.10.1 – 192.168.10.62

Step 3: Three Computer IP Addresses
Computer 1: 192.168.10.2
Computer 2: 192.168.10.3
Computer 3: 192.168.10.4

Step 4: Default Gateway
192.168.10.1

Finance LAN (192.168.10.64/26)

Step 1: Network and Broadcast Address
Network Address: 192.168.10.64
Broadcast Address: 192.168.10.127

Step 2: Host IP Range
192.168.10.65 – 192.168.10.126

Step 3: Three Computer IP Addresses

Computer 1: 192.168.10.66
Computer 2: 192.168.10.67
Computer 3: 192.168.10.68

Step 4: Default Gateway
192.168.10.65